Introduction to `transclust`

Extended on real world data

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This tutorial provides a brief introduction to the `transclust` package, which is designed for analysis of transmission clusters using a combination of whole genome sequencing (WGS) data and epidemiological information. In epidemiology, transmission clusters are groups of genetically similar pathogen isolates that are likely to have been transmitted from a common source. The package includes functions for clustering, visualization, and statistical analysis of transmission clusters. To get started, we will first install and load the package:

```
devtools::install_github("theabhirath/transclust")
library(transclust)
```

We will also load the `ape` package, which is used for phylogenetic analysis with the package and in the vignette, and `tidyverse`, `ggaalign` `paletteer` for data manipulation and visualization.

```
library(ape)
library(tidyverse)
library(ggalign)
library(paletteer)
```

Loading and preparing data

The `transclust` package comes with a sample dataset that can be used for demonstration purposes. The dataset contains epidemiological information for a set of isolates. To load the dataset, we can use the following command:

```
load(system.file("extdata", "example.Rdata", package = "transclust"))
```

We then read in the sequence files, which have been prepared using the procedure described in Hawken et al. (2022). The dataset itself provides one set of example sequences, and we use a few additional sequences for demonstration. We first read in the example sequence from the package using `ape`'s `read.dna` function:

```
dna_aln <- read.dna(system.file("extdata", "example.fasta", package = "transclust"), form
```

Now that the sequence file is loaded, we can extract the variable positions in the alignment. The `dna_var_pos` variable contains a logical vector indicating which columns in the alignment are variable. The `dna_pt_labels` variable contains the labels for the sequences in the alignment, and `trace_mat` is a

matrix containing the trace data for the isolates. The valid set of labels is determined by checking which labels in the `dna_pt_labels` variable are present in the `trace_mat` matrix.

```
# Get all variable positions in the alignment
var_pos <- apply(dna_aln, 2, function(x) sum(x == x[1]) < nrow(dna_aln))
# Only keep those labels that are in the trace matrix
valid_labels <- dna_pt_labels[labels(dna_aln)] %in% colnames(trace_mat)
```

We can then subset the alignment to include only the variable positions from the isolates with valid labels:

```
dna_var <- dna_aln[valid_labels, var_pos]
```

We then use the helper function `get_snp_dist_matrix()` to calculate the SNP distance matrix for these isolates:

```
snp_dist <- get_snp_dist_matrix(dna_var)
```

The `snp_dist` variable contains the SNP distance matrix, which is a square matrix where each entry represents the number of SNP differences between two isolates. The diagonal entries are all zero, as they represent the distance between an isolate and itself.

Clustering isolates using a hard SNP threshold

Now that we've loaded our data, we can start clustering the isolates. First, we want to use a hard SNP threshold to cluster the isolates. We first convert the SNP distance matrix into a phylogenetic tree. For the purposes of this analysis, we are going to construct a phylogenetic tree using the complete linkage method. The `hclust()` function from the `stats` package can be used to perform hierarchical clustering on the SNP distance matrix, which can then be converted into a phylogenetic tree using the `as.phylo()` function from the `ape` package.

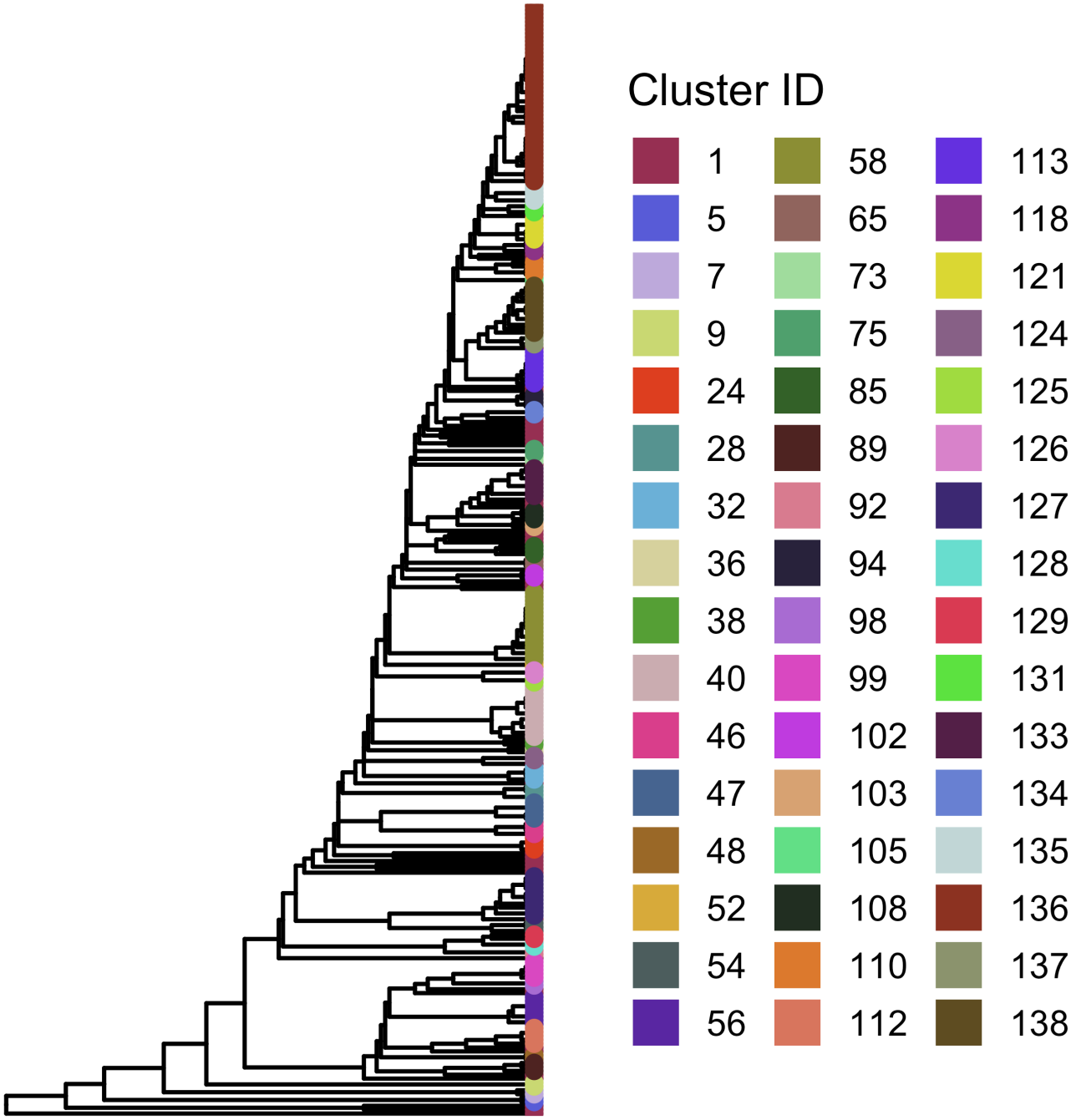
```
# Perform hierarchical clustering based on SNP distance matrix
snp_hclust <- hclust(as.dist(snp_dist))
# Convert hierarchical clustering into a phylogenetic tree using the complete linkage met
phylo_tree <- as.phylo(snp_hclust)
```

The `get_tn_clusters_snp_thresh()` function takes the SNP distance matrix and a threshold value as input and returns a list of clusters. We use a threshold of 10 SNPs for this example:

```
clusters_snp <- get_tn_clusters_snp_thresh(snp_hclust, phylo_tree, 10)
```

We can then visualize the clusters using the `plot_clusters_phylo()` function:

```
plot_clusters_phylo(clusters_snp, phylo_tree)
```



Clustering isolates using a threshold-free approach

In addition to the hard SNP threshold approach, we can also use a threshold-free approach to cluster the isolates. The package implements the method described in Hawken et al. (2022) in the `get_tn_clusters_sv_index()` function. The method uses a threshold-free approach to identify clusters based on the structure of the phylogenetic tree. For this function, we first need to generate a parsimony tree using the `get_phylo_tree()` function:

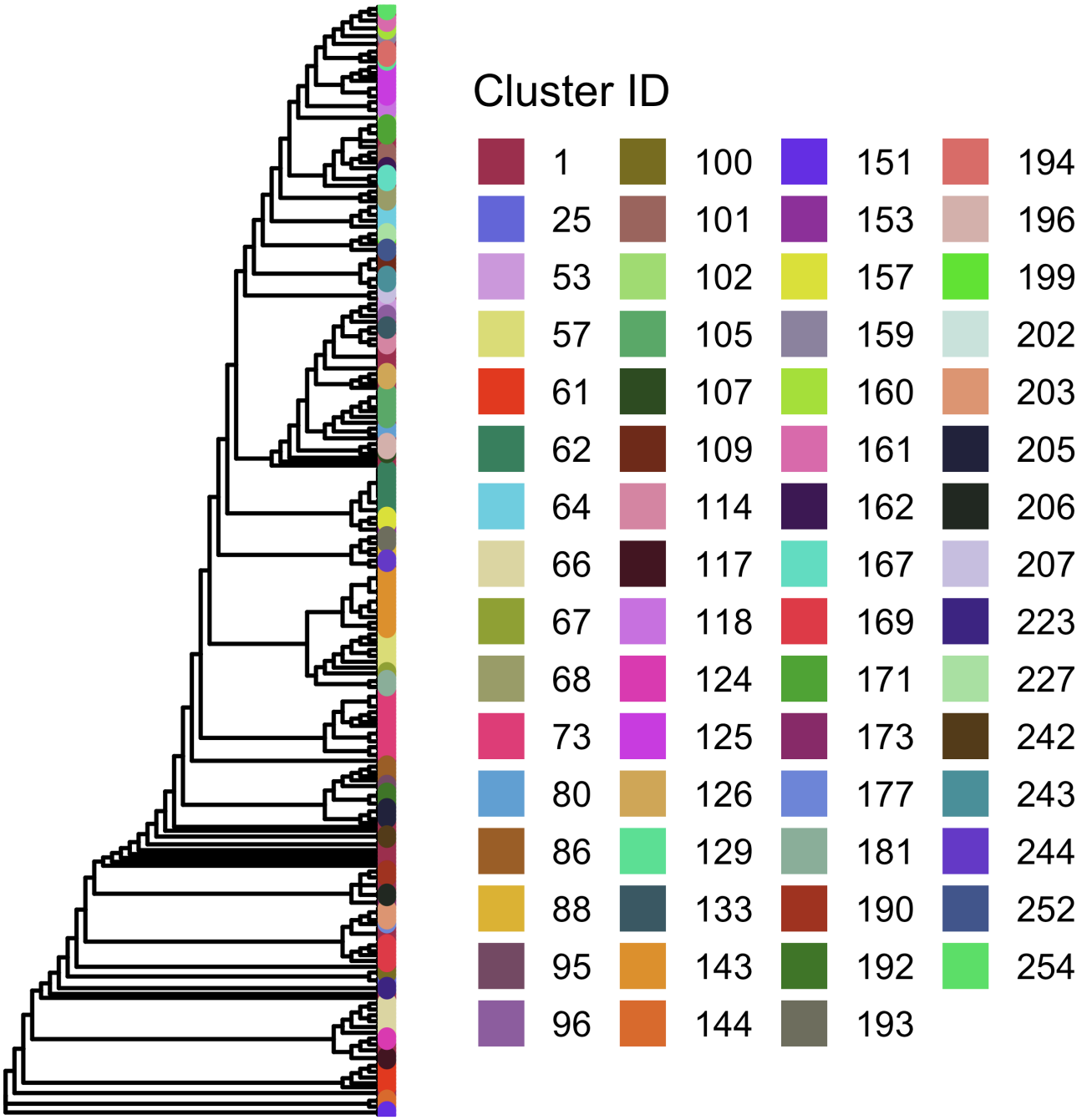
```
phylo_tree_pars <- get_phylo_tree(dna_var, snp_dist, "pars")
```

We can now use the `get_tn_clusters_sv_index()` function to identify clusters based on the parsimony tree. This function takes in a few more inputs than the SNP threshold function:

```
clusters_sv <- get_tn_clusters_sv_index(
    dna_var, snp_dist, ip_seqs_3days, ip_seqs,
    dna_pt_labels, dates, phylo_tree_pars
)
```

The `clusters_sv` object contains the clusters identified using the threshold-free approach. We can visualize these clusters on the parsimony tree:

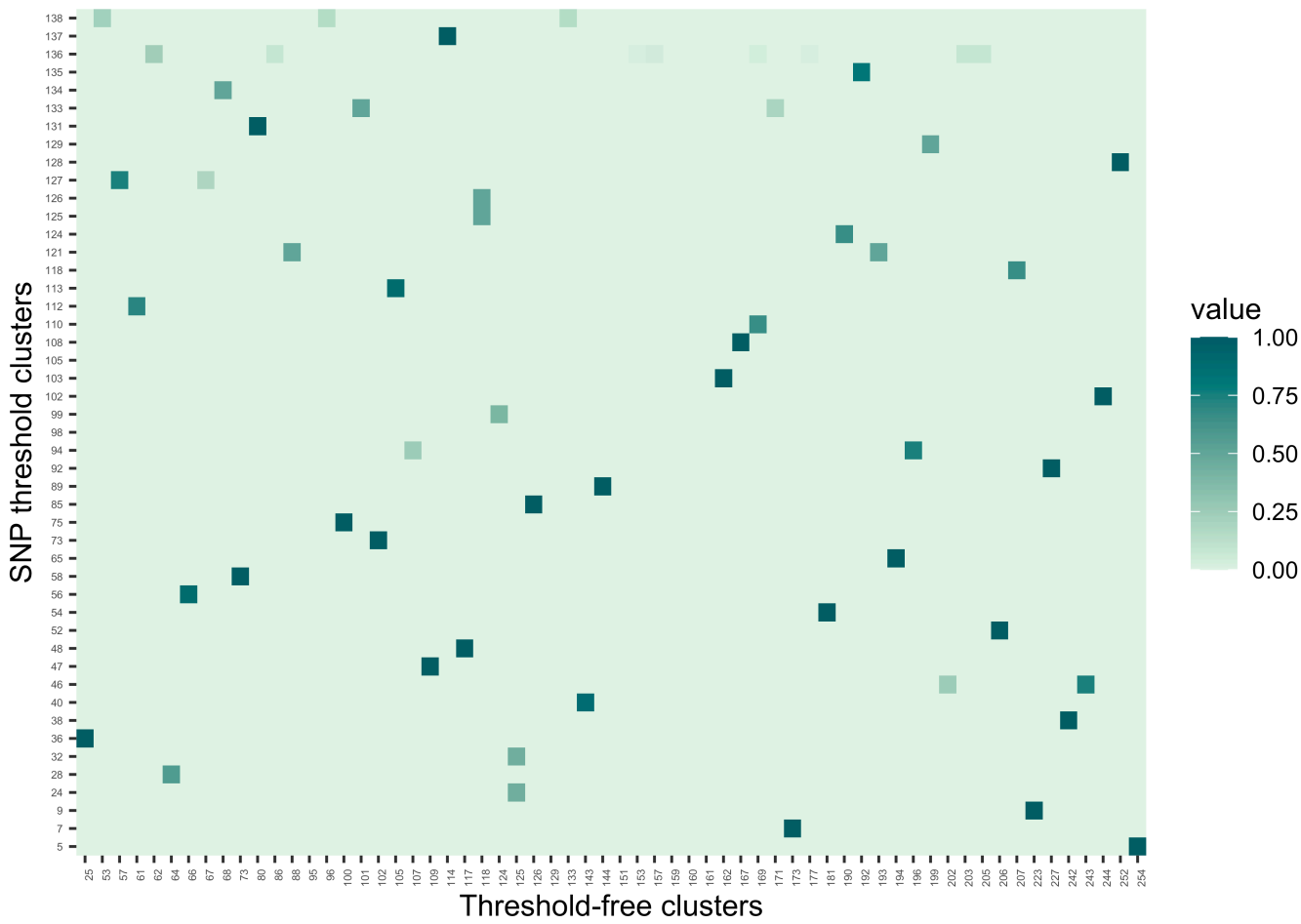
```
plot_clusters_phylo(clusters_sv, phylo_tree_pars)
```



Comparing clusters

These two trees look quite different. We can compare the clusters generated using the two methods using the `compare_clusters()` function, which generates a heatmap showing the overlap between the clusters by indicating the proportion of isolates in each cluster that are also present in the clusters generated by the other method:

```
compare_clusters(clusters_snp, clusters_sv) +
    # change the color palette
    scale_fill_paletteer_c("grDevices::Mint", direction = -1) +
    # add x and y axis labels
    xlab("Threshold-free clusters") +
    ylab("SNP threshold clusters") +
    # center the title and rotate x axis labels
    theme(
        plot.title = element_text(hjust = 0.5),
        axis.text.x = element_text(angle = 90, hjust = 1, size = 4),
        axis.text.y = element_text(size = 4)
    )
```



Adding more sequence data

We will now demonstrate the clustering using an expanded dataset. The workflow does not change at all, but we will use a different set of sequences:

```
# Read in the additional sequence data
dna_aln_1 <- read.dna("data/ST258.fasta", format = "fasta")
dna_aln_2 <- read.dna("data/ST258_2.fasta", format = "fasta")
# Get all variable positions in the alignment
var_pos_1 <- apply(dna_aln_1, 2, function(x) sum(x == x[1]) < nrow(dna_aln_1))
var_pos_2 <- apply(dna_aln_2, 2, function(x) sum(x == x[1]) < nrow(dna_aln_2))
# Only keep those labels that are in the trace matrix
valid_labels_1 <- dna_pt_labels[labels(dna_aln_1)] %in% colnames(trace_mat)
valid_labels_2 <- dna_pt_labels[labels(dna_aln_2)] %in% colnames(trace_mat)
# Subset the alignment to include only the variable positions from the isolates with vali
dna_var_1 <- dna_aln_1[valid_labels_1, var_pos_1]
dna_var_2 <- dna_aln_2[valid_labels_2, var_pos_2]
# Get SNP distance matrix
snp_dist_1 <- get_snp_dist_matrix(dna_var_1)
snp_dist_2 <- get_snp_dist_matrix(dna_var_2)
```

These sequences are from the same sequence type, ST258, but are samples taken from the patients at different times than the example dataset. We can use the same functions as before to generate the transmission clusters for both these sets of sequences using the shared variant method. Note that the epidemiological information for these sequences is the same because they are taken from the same hospital unit:

```
# Generate a parsimony tree
phylo_tree_pars_1 <- get_phylo_tree(dna_var_1, snp_dist_1, "pars")
phylo_tree_pars_2 <- get_phylo_tree(dna_var_2, snp_dist_2, "pars")
# Get the clusters using the shared variant method
clusters_sv_1 <- get_tn_clusters_sv_index(
    dna_var_1, snp_dist_1, ip_seqs_3days, ip_seqs,
    dna_pt_labels, dates, phylo_tree_pars_1
)
clusters_sv_2 <- get_tn_clusters_sv_index(
    dna_var_2, snp_dist_2, ip_seqs_3days, ip_seqs,
    dna_pt_labels, dates, phylo_tree_pars_2
)
```

Some statistical analysis of the clustering results

The `transclust` package also provides functions for statistical analysis of the clustering results. For example, we can use the `intra_cluster_genetic_var_analysis()` function to calculate the genetic

variation within each cluster:

```
mut_var_df <- intra_cluster_genetic_var_analysis(clusters_sv, dna_aln, var_pos)
```

This dataframe has columns for the total number of variable sites, and also the proportions of each of the six types of base substitutions present in each cluster. But this isn't very intuitive to just look at, so we can plot this. First, we drop any possible NA values. For this plot, we only want to compare relative frequencies of the different types of base substitutions, so we also exclude the first column - the total number of variable sites. We further subset the data frame to only include the first few rows for clarity in the plot:

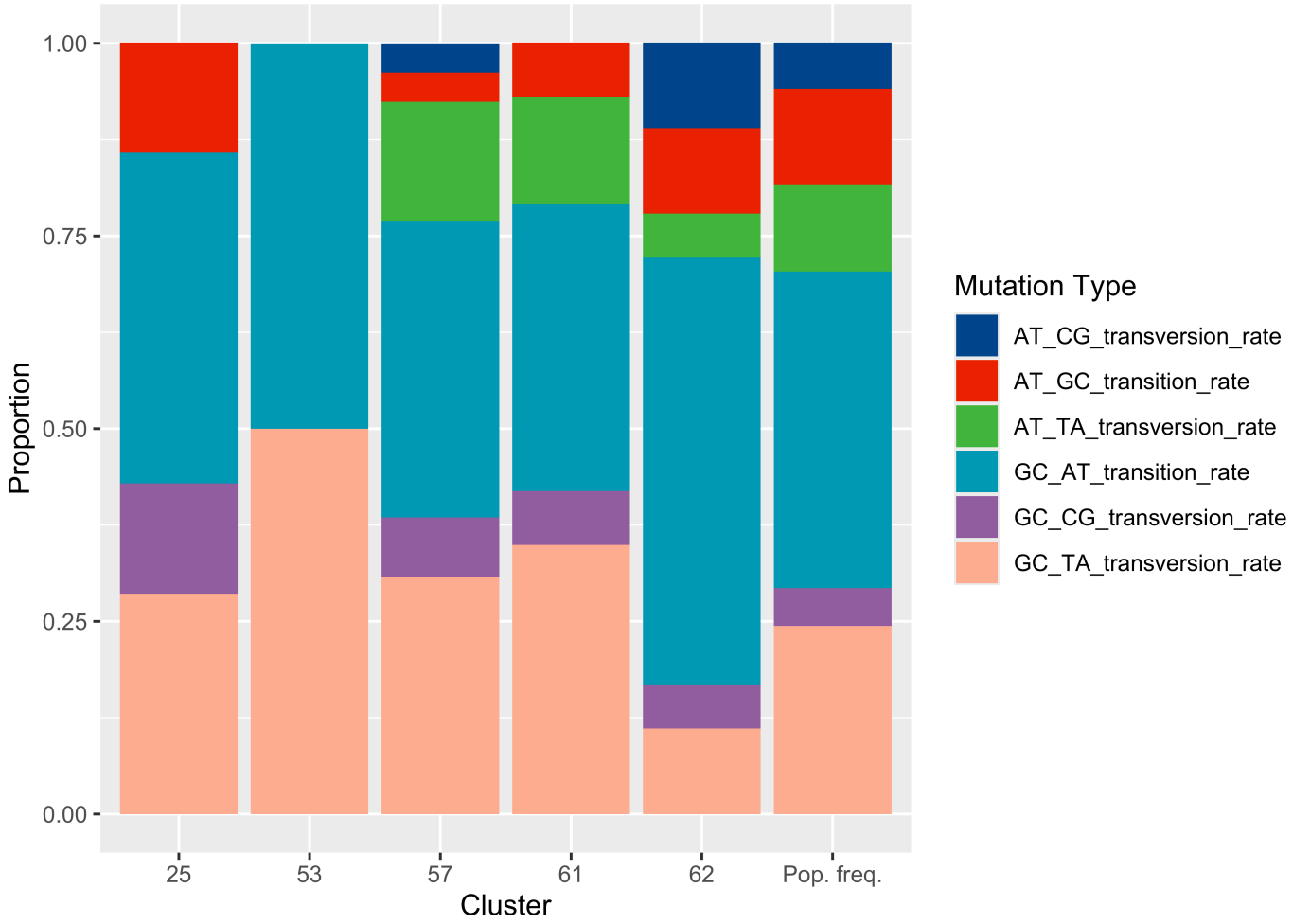
```
mut_var_df <- head(na.omit(mut_var_df)[, -1])
```

Then we can plot the proportions of each type of base substitution in each cluster. For this, the `ggplot2` package is used to create a stacked bar plot:

```
# Add column for cluster using rownames before pivoting
mut_var_df$cluster <- row.names(mut_var_df)

# Convert the data frame to long format for ggplot
mut_var_df_long <- pivot_longer(
    mut_var_df,
    cols = -cluster,
    names_to = "mutation_type",
    values_to = "proportion"
)

# Create the stacked bar plot
ggplot(mut_var_df_long, aes(x = cluster, y = proportion, fill = mutation_type)) +
    geom_bar(stat = "identity") +
    labs(x = "Cluster", y = "Proportion", fill = "Mutation Type") +
    scale_fill_paletteer_d("ggsci::lanonc_lancet")
```



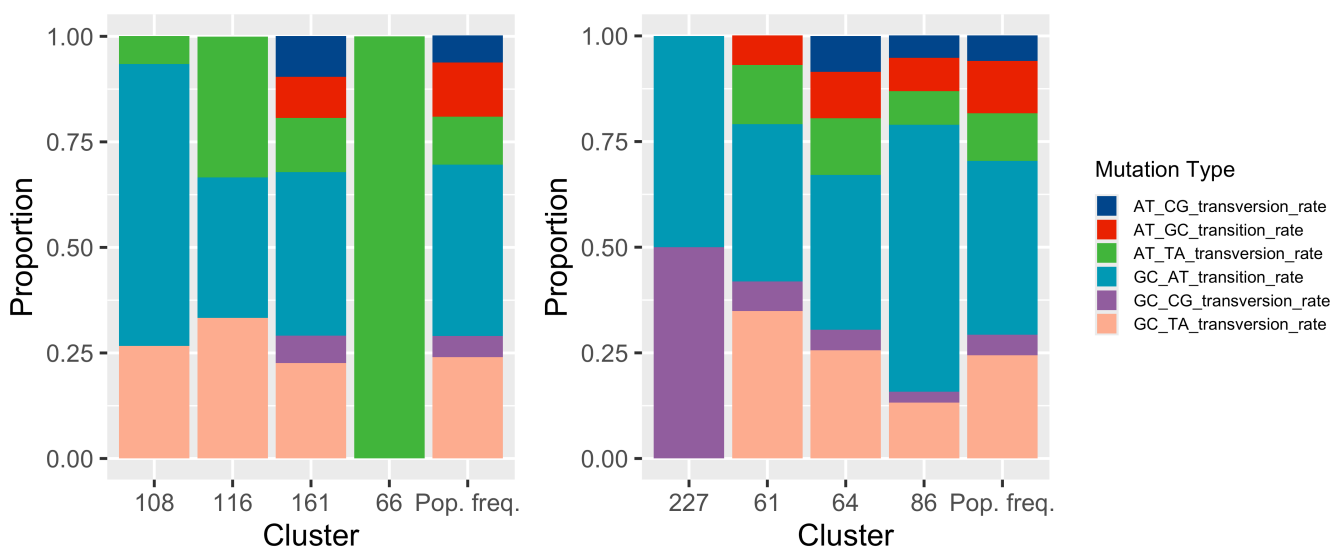
We can clearly see very specific mutational signatures emerge in the clusters when compared to the population as a whole. Now, we can extend this to the larger dataset to see the mutational signatures there:

```
# Add column for cluster using rownames before pivoting
mut_var_df_1 <- intra_cluster_genetic_var_analysis(clusters_sv_1, dna_aln_1, var_pos_1)
mut_var_df_1 <- na.omit(mut_var_df_1)[, -1] |> slice_sample(n = 5)
mut_var_df_1$cluster <- row.names(mut_var_df_1)
mut_var_df_2 <- intra_cluster_genetic_var_analysis(clusters_sv_2, dna_aln_2, var_pos_2)
mut_var_df_2 <- na.omit(mut_var_df_2)[, -1] |> slice_sample(n = 5)
mut_var_df_2$cluster <- row.names(mut_var_df_2)

# Convert the data frame to long format for ggplot
mut_var_df_long_1 <- pivot_longer(
    mut_var_df_1,
    cols = -cluster,
    names_to = "mutation_type",
    values_to = "proportion"
)
mut_var_df_long_2 <- pivot_longer(
    mut_var_df_2,
    cols = -cluster,
```

```r
    names_to = "mutation_type",
    values_to = "proportion"
)

# Create two stacked bar plots side by side
p1 <- ggplot(mut_var_df_long_1, aes(x = cluster, y = proportion, fill = mutation_type)) +
    geom_bar(stat = "identity") +
    labs(x = "Cluster", y = "Proportion", fill = "Mutation Type") +
    scale_fill_paletteer_d("ggsci::lanonc_lancet") +
    theme(legend.position = "none")
p2 <- ggplot(mut_var_df_long_2, aes(x = cluster, y = proportion, fill = mutation_type)) +
    geom_bar(stat = "identity") +
    labs(x = "Cluster", y = "Proportion", fill = "Mutation Type") +
    scale_fill_paletteer_d("ggsci::lanonc_lancet") +
    theme(legend.position = "none")
# Combine the two plots
combined_plot <- p1 + p2 +
    theme(
        legend.position = "right",
        legend.title = element_text(size = 8),
        legend.text = element_text(size = 6),
        legend.key.size = unit(0.3, "cm"),
    )
combined_plot
```



We can very clearly see the mutational signature differences between the different clusters, but the population signatures don't change much since these sequences belong to the same sequence type, i.e. ST258.

Conclusion

In this tutorial, we have provided a brief introduction to the `transclust` package and demonstrated how to use it for clustering transmission clusters using whole genome sequencing data and epidemiological information. We have shown how to load and prepare the data, perform clustering using both SNP-threshold and threshold-free approaches, visualize the clusters, and perform some basic statistical analysis of the clustering results. The package provides a powerful set of tools for analyzing transmission clusters and can be used to gain insights into the dynamics of pathogen transmission by studying transmission clusters.

References

Hawken, Shawn E, Rachel D Yelin, Karen Lolans, Ali Pirani, Robert A Weinstein, Michael Y Lin, Mary K Hayden, and Evan S Snitkin. 2022. "Threshold-Free Genomic Cluster Detection to Track Transmission Pathways in Health-Care Settings: A Genomic Epidemiology Analysis." *The Lancet Microbe* 3 (9): e652–62. https://doi.org/10.1016/s2666-5247(22)00115-x.